Few-Shot Lifelong Hyperspectral Image Classification With Dual-Branch Contrast and Geometric Alignment

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***Abstract*—Hyperspectral image (HSI) classification in few-shot lifelong learning faces dual challenges: catastrophic forgetting from inaccessible historical training samples and overfitting due to novel-class sample scarcity. To address these challenges, we propose HyperDual, a novel framework for few-shot lifelong HSI classification. HyperDual integrates three key components: the dual-branch contrastive optimization (DBCO) module, the stability-diversity replay (SDR) mechanism, and the aligned-distilled feature consistency (ADFC) module. Based on the generalizability and transferability of shallow neural networks and the robustness of contrastive learning, the DBCO module jointly employs a self-supervised contrastive (SupCon) learning branch to intermediate-layer features from the backbone network to enhance generalization capability, and a SupCon learning branch to the output features from the projection layer to improve interclass separability, providing a unified feature representation for both base and incremental classes after freezing the backbone. The SDR mechanism mitigates catastrophic forgetting and feature overlap between old and new classes by jointly replaying deterministic prototypes and diverse Gaussian-sampled pseudo-intermediate features. Motivated by the neural collapse (NC) theory, where the penultimate-layer features converge to class prototypes, forming a simplex equiangular tight frame (ETF), the ADFC module aligns all output features with the corresponding weights of the preassigned ETF classifier while maintaining distribution consistency between the current and previous projection layers. Extensive experiments on the Indian Pines, Houston 2013, and Pavia University datasets demonstrate that HyperDual outperforms state-of-the-art methods. Code: https://github.com/lzlsxs/HyperDual**

***Index Terms*—Contrastive learning, few-shot, geometric alignment, hyperspectral image (HSI) classification, lifelong learning.**

Received 18 June 2025; revised 28 July 2025 and 16 August 2025; accepted 18 August 2025. Date of publication 21 August 2025; date of current version 3 September 2025. This work was supported in part by the National Natural Science Foundation of China under Grant 62425109, Grant U22B2014, and Grant 72403085; in part by the Science and Technology Plan Project Fund of Hunan Province under Grant 2022RSC3064; and in part by Hunan Provincial Natural Science Foundation of China under Grant 2023JJ40024 and Grant 2025JJ40059. *(Corresponding authors: Wei Bao; Jun Yue.)*

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Digital Object Identifier 10.1109/TGRS.2025.3601529

I. Introduction

HYPERSPECTRAL images (HSIs), acquired by high-resolution imaging spectrometers, comprise hundreds of contiguous, narrow spectral bands. Each pixel in HSIs corresponds to a spectral signature that encodes the reflectance properties of distinct surface materials or land-cover categories. HSI classification, a process assigning the correct land-cover category to each pixel by resolving subtle interclass spectral variations, is critical for precision agriculture [1], military reconnaissance [2], environmental monitoring [3], and mineral exploration [4], among various other applications.

Deep learning-based HSI classification has achieved remarkable progress through successive architectural innovations [5], [6], [7], [8], [9], [10]. However, prevailing methodologies predominantly operate under closed-set assumptions, relying on static training datasets with fixed land-cover categories. In real-world scenarios, land-cover taxonomies continuously evolve, while privacy regulations and storage limitations often preclude access to historical training data. Consequently, conventional approaches suffer from catastrophic forgetting, a phenomenon in which models abruptly discard prior knowledge when learning new categories [11], [12]. Inspired by human cognitive plasticity, lifelong learning (also termed incremental or continual learning) has been introduced to equip neural networks with autonomous adaptability to novel categories while maintaining stability in historical categories [13], [14]. Exemplar-replay lifelong learning methods [15], [16], [17], which mitigate catastrophic forgetting by storing representative samples or generators from previous tasks, are considered highly effective strategies for lifelong learning. However, these methods still face practical constraints such as data privacy risks and storage overhead. Exemplar-free lifelong learning methods eliminate the dependency on stored samples through various mechanisms, such as distillation-based regularization [18], [19], [20], prototype-based alignment [21], [22], pseudo-sample-based replay [23], [24], [25], and foundation model-based methods [26], [27], [28].

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While lifelong learning approaches have partially alleviated the challenge of catastrophic forgetting, these methods assume the availability of sufficient training data for new classes [29], [30]. This assumption remains a critical bottleneck in real-world scenarios, particularly in domains like HSI classification, where collecting large-scale samples for novel classes is prohibitively expensive, and data annotation remains labor-intensive and costly. When limited training samples are available for new classes, models must simultaneously address catastrophic forgetting and overcome the additional challenge of overfitting [31]. To tackle this, few-shot lifelong learning has been proposed as a research direction and has gained significant attention [32].

Existing few-shot lifelong learning methods typically adopt a base-to-incremental paradigm where feature extractors are frozen after initial training [33], [34]. These approaches can be broadly categorized into data replay-based methods, distillation-based methods, dynamic network structure-based methods, metalearning-based methods, foundation model-based methods, and feature space-based methods [31], [35].

Data replay-based methods [36], [37], [38] store representative samples or generators from previous sessions, allowing historical samples (either real or synthetic) to be replayed during incremental sessions. While effective in mitigating catastrophic forgetting, their practicality is constrained by storage limits, privacy issues, and generator training complexity. Distillation-based methods [39], [40], [41] maintain feature consistency between current and previous models, jointly mitigating catastrophic forgetting while addressing challenges in data imbalance and overfitting. However, relying solely on knowledge distillation [42] yields limited performance improvements. Dynamic network structure-based methods [29], [43], [44], [45] employ structures such as neural gas networks, graph attention networks [46], and dynamic neural networks to enable real-time structural adaptation based on input data characteristics, which enhances generalization and reduces overfitting risks. However, architectural complexity and limited generalizability across diverse tasks significantly hinder their adoption. Metalearning-based methods [47], [48] optimize models through pseudo-incremental tasks, formalizing knowledge retention and adaptation as a metaoptimization objective. However, when new session distributions diverge from the base session, performance degrades significantly. Foundation models exhibit efficient domain generalization [49]. Foundation model-based methods [50], [51], [52] leverage pretrained vision transformers (ViTs) or multimodal foundation models (e.g., CLIP [53]) to achieve few-shot adaptation to new tasks. However, these methods require substantial computational resources and depend on the availability of pretrained models. Feature space-based methods [30], [54], [55], [56] address catastrophic forgetting and overfitting through two principles: mitigating feature distribution shifts of old classes and maximizing interclass separation while compressing intraclass variances in the embedding space.

Recent studies [33], [34] have leveraged fixed optimal geometric classifiers (e.g., equiangular tight frame (ETF) classifiers [57]) for few-shot lifelong learning. By enforcing persistent cosine similarity alignment between adapted features and predefined classifier weights, these methods theoretically ensure feature-classifier alignment. While these approaches have demonstrated promising results and attracted considerable attention, the field of few-shot lifelong learning for HSI classification still faces the following critical challenges.

1) Conventional approaches train classification networks by optimizing classification loss functions (e.g., cross-entropy) during the base session, achieving high accuracy on base classes. However, the widely adopted practice of freezing the backbone network inherently limits the model's feature representation for novel classes in incremental sessions, ultimately leading to entanglement in the embedding space.
2) Recent attempts employ contrastive learning to enhance the robustness and generalization ability of the classification network for subsequent incremental sessions. However, updating components of the classification network (e.g., the projection layer) paradoxically leads to reduced robustness and generalization capability during these sessions.
3) In incremental sessions, the backbone and fixed classifier remain fixed. The goal is to optimize the projection layer. However, lacking training samples from previous classes, the projection layer overfits to the few-shot data of current classes, disrupting the mapping between intermediate features of previous classes and their corresponding ETF classifier weights.

To address these challenges, we propose HyperDual, a feature space-based few-shot lifelong learning framework for HSI classification, which introduces three innovative components: the dual-branch contrastive optimization (DBCO) module, the stability-diversity replay (SDR) mechanism, and the aligned-distilled feature consistency (ADFC) module. The key contributions of this work are summarized as follows.

1) Based on the generalizability and transferability of shallow neural networks [58] and the robustness of contrastive learning [59], [60], [61], our DBCO module resolves the inherent conflict between backbone freezing and novel-class adaptability. It jointly applies a self-supervised contrastive (SupCon) branch to enhance the backbone's generalization, and a SupCon branch to improve the interclass separability in the projection layer, providing unified feature representations for both base and incremental classes after freezing the backbone.
2) To mitigate catastrophic forgetting and address feature overlap between old and new classes, SDR jointly replays deterministic prototypes and diverse pseudo-intermediate features, enforcing that the projection layer preserves mappings between intermediate features and their corresponding classifier weights for old classes, when learning such mappings for new classes.
3) The ADFC module minimizes the angular deviation between output features and their corresponding weight vectors in the preassigned ETF classifier, while employing distillation loss to align the feature distributions

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of the current and previous projection layers, thereby achieving optimal geometric alignment.

The remainder of this article is organized as follows. Section II introduces related work on neural collapse (NC) and contrastive learning. Section III details HyperDual, our novel few-shot lifelong learning framework tailored for HSI classification. Section IV comprehensively describes the HSI datasets, implementation configurations, experimental results, and analysis. Section V concludes the study and suggests future work.

II. Related Work

A. Neural Collapse

NC describes a geometric phenomenon emerging in deep neural networks during terminal phase training (with training error approaching zero) on balanced datasets [57]. This phenomenon manifests through the formation of a simplex ETF between the penultimate-layer features and the classifier weights for classification tasks. Under NC, the variance of feature distributions within each class in the penultimate layer diminishes to zero, causing all features to collapse to their class-specific mean prototypes. Simultaneously, all class means converge to a simplex ETF frame, a symmetric geometric configuration that maximizes interclass separation. The classifier weight vectors also align with the simplex ETF, enabling classification through a nearest-class-mean decision rule in the feature space. In this state, the simplex ETF comprises K vectors $\mathbf{W}_{\text{ETF}} = [\boldsymbol{w}_1, \ldots, \boldsymbol{w}_K] \in \mathbb{R}^{d\times K}$, which satisfy

$$\boldsymbol{w}_i^{\top}\boldsymbol{w}_j = \frac{K}{K-1}\delta_{i,j} - \frac{1}{K-1} \quad \forall i,\ j \in [1, K] \tag{1}$$

where K is the number of classes, $d \geq K-1$, $\delta_{i,j} = 1$ when $i = j$, and 0 otherwise.

Yang et al. [33] first introduced the NC theory into few-shot lifelong learning, proposing the NC-FSCIL method. The NC-FSCIL framework employs a preassigned ETF classifier during the initial training phase. Throughout training, it maintains the classifier fixed while constraining each sample's penultimate-layer features to be parallel to or aligned with the corresponding class-specific weight vector. This mechanism ensures that the feature means of both old and new classes during incremental phases consistently converge to the preassigned ETF classifier weights, thereby preserving the optimal interclass separability embedded in the ETF geometric configuration. Dang et al. [62] addressed the challenge of intraclass diversity degradation and intertask representation overlap in NC-FSCIL by directly incorporating NC constraints into the loss function. This approach combines hardness-softness plasticity and hardness-softness distillation to alleviate catastrophic forgetting while learning new knowledge. Seo et al. [63] proposed preparatory data training and residual correction for online lifelong learning to induce NC to form a simplex ETF structure in the representation space, enabling the model to adapt incrementally via single-epoch training and better fit the streaming data. Ahmed et al. [34] proposed OrCo, which replaces the fixed ETF classifier with a fixed classifier whose weight vectors are mutually orthogonal. By synergizing the orthogonal feature geometry with contrastive learning, OrCo enhances the generalizability of the feature space while mitigating data scarcity in few-shot lifelong learning scenarios.

B. Contrastive Loss

Contrastive learning establishes a discriminative embedding space that minimizes intraclass variance while maximizing interclass separation, proving particularly beneficial for downstream tasks requiring transferable representations, such as classification and lifelong learning. As described in [59], Bromley et al. [64] introduced the contrastive loss, subsequently formalized by Chopra et al. [65] and Hadsell et al. [66]. Mainstream contrastive learning methods encompass the InfoNCE loss proposed by van den Oord et al. [67], the SimCLR framework developed by Chen et al. [68], and the SupCon loss proposed by Khosla et al. [69].

Recent advances in contrastive learning demonstrate its capability to learn transferable visual representations that exhibit superior generalizability to unseen tasks compared with cross-entropy-based methods constrained by task-specific supervision. Cha et al. [60] first applied contrastive learning to lifelong learning, proposed the contrastive continual learning (Co^2L) framework, and demonstrated that contrastively trained representations suffer less catastrophic forgetting than those trained with cross-entropy loss. Fini et al. [61] established a generalized methodology for transforming self-supervised loss functions into plasticity-preserving distillation mechanisms, and demonstrated that self-SupCon learning can learn robust representations against catastrophic forgetting in continual learning. To address representation bias, Li et al. [70] developed importance sampling strategies to align contrastive embeddings between online and offline training phases. Mai et al. [71] introduced intraclass compactness and interclass separation objectives during replay-based training to mitigate catastrophic forgetting. Wen et al. [59] further enhanced Co^2L through dynamic coefficient optimization for distillation losses. Most recently, Dang et al. [62] proposed focal NC contrastive (FNC^2) learning, which implements hardness-softness plasticity. While these methods demonstrate progressive improvements, their predominant reliance on exemplar replay buffers imposes practical deployment constraints due to memory storage requirements and privacy concerns.

III. Proposed Method

A. Problem Setup

In few-shot lifelong learning for HSI classification, the model incrementally acquires knowledge from a sequence of sessions $\{\mathcal{D}_0, \ldots, \mathcal{D}_T\}$. Each session $\mathcal{D}_t = \{\mathcal{X}_t, \mathcal{Y}_t\}$ consists of input samples $\mathcal{X}_t = \{\boldsymbol{x}_i\}_{i=1}^{n_t}$ and their corresponding labels $\mathcal{Y}_t = \{y_i \in \mathcal{C}_t\}_{i=1}^{n_t}$, where $\mathcal{C}_t$ represents the class set, $|\mathcal{C}_t|$ is the number of classes, and n_t denotes the total number of samples in session t. The base session $\mathcal{D}_0$ contains a large-scale dataset with numerous classes and sufficient training samples per class. Subsequent sessions $\mathcal{D}_t$ ($t > 0$), termed incremental sessions, introduce N novel classes, with each class restricted to only K training samples, forming an N-way

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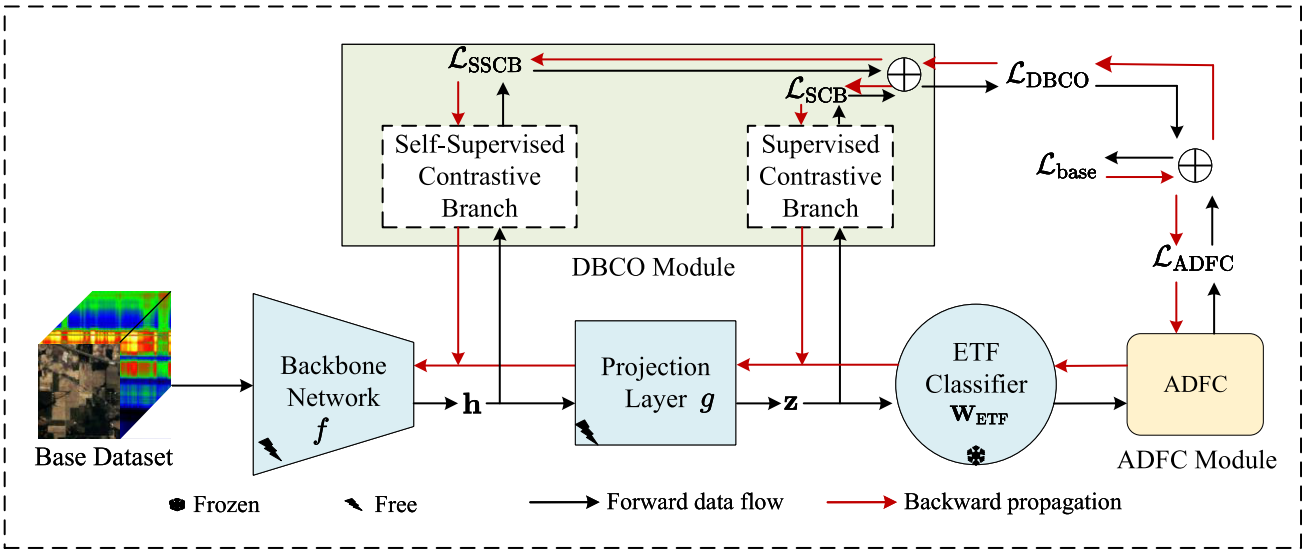


Fig. 1. Flowchart of base session training. The classification network comprises: 1) a trainable backbone network f that extracts intermediate features $\boldsymbol{h}$; 2) a trainable projection layer g that projects $\boldsymbol{h}$ into output features $\boldsymbol{z}$; and 3) a preassigned ETF classifier $\mathbf{W}_{\text{ETF}}$. The DBCO module jointly applies a self-SupCon learning branch to the intermediate features $\boldsymbol{h}$ to enhance generalization capability, and a SupCon learning branch to the output features $\boldsymbol{z}$ to improve interclass separability. Furthermore, the ADFC module aligns the output features $\boldsymbol{z}$ with corresponding classifier weight vectors, establishing an optimal geometric structure that maximizes interclass separability and intraclass compactness.

K-shot learning scenario. Class sets are strictly disjoint across sessions.

During each incremental session, the model exclusively accesses the current session's limited samples ($N \times K$ instances) and cannot revisit raw data from historical sessions, emulating practical constraints such as data privacy protocols or storage limitations. Despite these restrictions, the model is required to retain comprehensive classification capability over the union of all previously encountered classes $\bigcup_{i=0}^{t} \mathcal{C}_i$.

B. Overview of the Proposed Framework

The proposed framework, as illustrated in Figs. 1 and 2, addresses catastrophic forgetting and few-shot overfitting through three synergistic components: the DBCO module, SDR mechanism, and ADFC module.

Following Yang et al. [33], the classification network comprises a backbone network f, a projection layer g, and a preassigned ETF classifier $\mathbf{W}_{\text{ETF}} \in \mathbb{R}^{d\times K}$. $\mathbf{W}_{\text{ETF}}$ remains frozen across all sessions. During the base session, both the backbone network f and the projection layer g are trainable, whereas in incremental sessions, only the projection layer g remains trainable while the backbone f is frozen. During forward propagation, a training sample $\boldsymbol{x} \in \mathbb{R}^{p\times w\times h}$ is first processed by the backbone network f to extract an intermediate feature $\boldsymbol{h} \in \mathbb{R}^d$

$$\boldsymbol{h} = f(\boldsymbol{x}). \tag{2}$$

The projection layer g projects the intermediate feature $\boldsymbol{h}$ into an output feature $\boldsymbol{z} \in \mathbb{R}^d$

$$\boldsymbol{z} = g(\boldsymbol{h}). \tag{3}$$

The output feature $\boldsymbol{z}$ is then classified by the fixed ETF classifier $\mathbf{W}_{\text{ETF}}$.

In the base session, the DBCO module jointly applies a self-SupCon learning branch to intermediate features from the backbone network to enhance its generalization capability, and a SupCon learning branch to output features from the projection layer to improve interclass separability. Furthermore, the ADFC module aligns output features of the projection layer with corresponding weight vectors of the ETF classifier, thereby achieving an optimal geometric structure that maximizes interclass separability and intraclass compactness.

In incremental sessions, the SDR mechanism mitigates catastrophic forgetting and resolves feature overlap between old and new classes by replaying deterministic prototypes and diverse pseudo-intermediate features, thereby preserving mappings between intermediate features and corresponding classifier weight vectors for old classes. Furthermore, the ADFC module ensures that all output features remain geometrically aligned with corresponding weight vectors of the ETF classifier, while enforcing consistency between the output distributions of the current projection layer and those of the previous projection layer.

C. Dual-Branch Contrastive Optimization

Shallow neural layers exhibit higher generalizability and transferability while deep layers show stronger task-specificity [58], whereas contrastively trained representations suffer less catastrophic forgetting than those trained with cross-entropy loss [60]. Moreover, self-SupCon learning obtains robust representations against catastrophic forgetting in continual learning [61]. These inherent properties motivate our DBCO module, as illustrated in Fig. 1.

1) Self-Supervised Branch for Generalizable Backbone Features: We enforce the cross-session generalizability of the backbone network through self-SupCon learning. Each input sample $\boldsymbol{x}_i$ is augmented via horizontal, vertical, and diagonal flips to generate multiple views. The positive set $\mathcal{P}_i$ includes all augmented views of the anchor sample; the negative set

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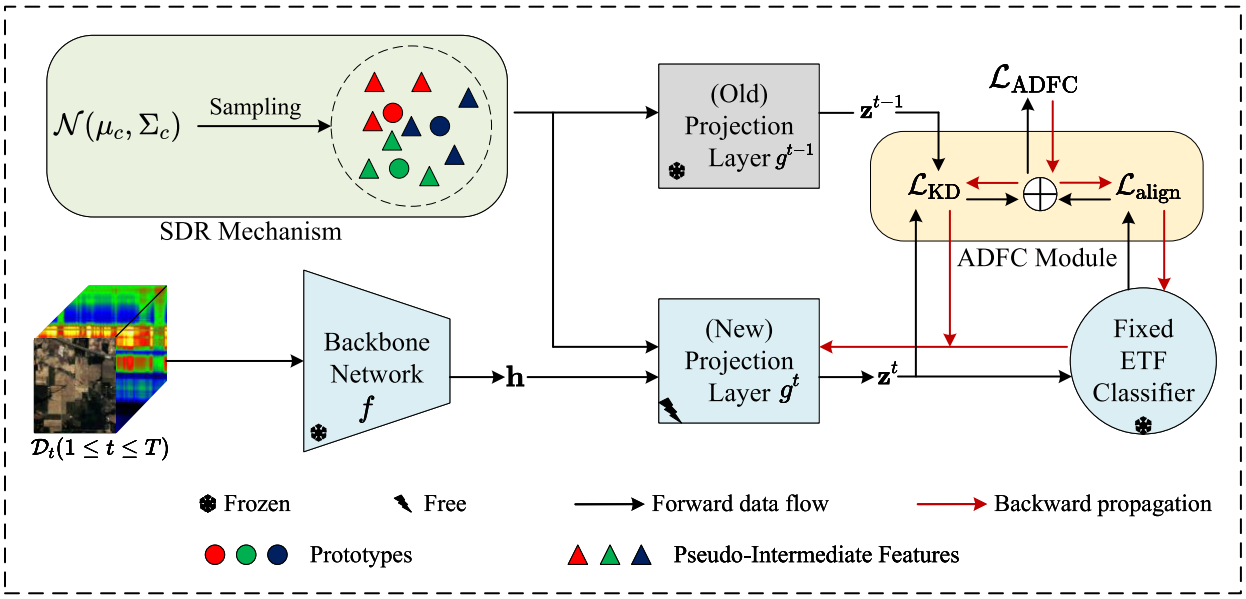


Fig. 2. Flowchart of incremental session t. In this session, only the projection layer g_t remains trainable while the backbone and the ETF classifier $\mathbf{W}_{\text{ETF}}$ are frozen. The real samples of the current session are processed through the backbone f and current projection layer g_t to generate output features. Simultaneously, the SDR mechanism synthesizes prototypes and pseudo-intermediate features for prior classes, which are then mapped to output features via the current projection layer g_t. The ADFC module constrains all output features to ensure alignment with the weight vectors of the ETF classifier $\mathbf{W}_{\text{ETF}}$, while distillation constraints align the feature distributions generated by g_t and previous projection layers g_{t-1}. By jointly replaying prototypes and pseudo-intermediate features, the current projection layer g_t preserves the feature-classifier mapping of old classes, while learning such mappings for new classes.

$\mathcal{N}_i$ comprises all other samples and their augmentations. By applying the InfoNCE loss [67] to intermediate features, the backbone learns robust representations, thereby improving its generalizability to unseen classes in incremental sessions. This branch constrains the backbone to capture generic low-level features while maintaining invariance to nuisance factors, thereby forming a stable representational baseline for incremental sessions. It can be formulated as follows:

$$\boldsymbol{h}_i = f(\boldsymbol{x}_i), \quad \boldsymbol{h}_j = f(\boldsymbol{x}_j) \tag{4}$$

$$\mathcal{L}_{\text{SSB}} = -\frac{1}{|\mathcal{D}_0| \cdot |\mathcal{P}_i|} \sum_{\boldsymbol{x}_i \in \mathcal{D}_0} \sum_{\boldsymbol{x}_j \in \mathcal{P}_i} \log \frac{\exp\left(\langle \boldsymbol{h}_i, \boldsymbol{h}_j \rangle / \tau\right)}{\sum_{\boldsymbol{x}_k \in \mathcal{A}_i} \exp\left(\langle \boldsymbol{h}_i, \boldsymbol{h}_k \rangle / \tau\right)} \tag{5}$$

where $\boldsymbol{h}_i \in \mathbb{R}^d$ and $\boldsymbol{h}_j \in \mathbb{R}^d$ are intermediate features, $\mathcal{D}_0$ denotes the base session training set, $\mathcal{A}_i = \mathcal{P}_i \cup \mathcal{N}_i$ represents the union of the positive set and the negative set, $\langle \cdot, \cdot \rangle$ represents the cosine similarity, and τ denotes the temperature factor.

2) Supervised Branch for Discriminative Embedding: The projection layer maps intermediate features from the backbone to discriminative output features. This branch enforces intraclass compactness and interclass separability by applying SupCon learning to output features, where the positive set $\mathcal{P}_i$ for an anchor $\boldsymbol{x}_i$ includes all samples of the same class and their augmentations, while the negative set $\mathcal{N}_i$ contains samples from all other classes (excluding the anchor's class) and their augmented instances. It can be formulated as follows:

$$\boldsymbol{z}_i = g(f(\boldsymbol{x}_i)), \quad \boldsymbol{z}_j = g(f(\boldsymbol{x}_j)) \tag{6}$$

$$d_{\text{pos}}(\boldsymbol{x}_i) = \frac{1}{|\mathcal{P}_i|} \sum_{\boldsymbol{x}_j \in \mathcal{P}_i} \left(1 - \langle \boldsymbol{z}_i, \boldsymbol{z}_j \rangle\right)^2 \tag{7}$$

$$d_{\text{neg}}(\boldsymbol{x}_i) = \frac{1}{|\mathcal{N}_i|} \sum_{\boldsymbol{x}_j \in \mathcal{N}_i} \left(1 - \langle \boldsymbol{z}_i, \boldsymbol{z}_j \rangle\right)^2 \tag{8}$$

$$\mathcal{L}_{\text{SB}} = \frac{1}{|\mathcal{D}_0|} \sum_{\boldsymbol{x}_i \in \mathcal{D}_0} \frac{d_{\text{pos}}(\boldsymbol{x}_i)}{d_{\text{pos}}(\boldsymbol{x}_i) + d_{\text{neg}}(\boldsymbol{x}_i)} \tag{9}$$

where $\boldsymbol{z}_i$ and $\boldsymbol{z}_j$ are the output features, d_{pos} represents the average distance from all samples in the positive set $\mathcal{P}_i$ for the anchor $\boldsymbol{x}_i$, d_{neg} denotes the average distance from all samples in the negative set $\mathcal{N}_i$ to the anchor $\boldsymbol{x}_i$, $\langle \cdot, \cdot \rangle$ represents the cosine similarity, and $\mathcal{D}_0$ denotes the base session training set.

The loss function for the DBCO module can be formulated as follows:

$$\mathcal{L}_{\text{DBCO}} = \lambda_1 \mathcal{L}_{\text{SSB}} + (1 - \lambda_1) \mathcal{L}_{\text{SB}} \tag{10}$$

where $\lambda_1 \in [0, 1]$ is a hyperparameter.

In addition, in the base session, we align output features of the projection layer with the corresponding weight vectors in the ETF classifier via ADFC to achieve a simplex ETF. The DBCO and ADFC modules jointly ensure transferable feature learning, intraclass contrastive clustering, and interclass separability, and geometric optimality, providing a unified feature representation for both base and incremental classes. Notably, since no previous model exists in the base session, the cross-session distribution distillation term in ADFC is naturally absent. Thus, the ADFC loss reduces to solely the feature-classifier geometric alignment term defined in (15).

The total loss function for the base session can be formulated as follows:

$$\mathcal{L}_{\text{base}} = \lambda_2 \mathcal{L}_{\text{DBCO}} + \mathcal{L}_{\text{align}}. \tag{11}$$

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D. Stable-Diverse Replay

To mitigate catastrophic forgetting and resolve feature overlap between old and new classes in incremental sessions, the proposed SDR mechanism introduces a SDR mechanism for spectral–spatial features to preserve discriminative interclass feature distributions.

In each incremental session t, for the new class $c \in \mathcal{C}_t$, the frozen backbone network extracts the spectral–spatial intermediate feature $\boldsymbol{h}_i^c \in \mathbb{R}^d$ from each sample $\boldsymbol{x}_i \in \mathcal{D}_t^c$, where $\mathcal{D}_t^c$ denotes the sample set of class c in session t. The SDR mechanism computes and stores the classwise mean $\boldsymbol{\mu}_c \in \mathbb{R}^d$ and covariance matrix $\boldsymbol{\Sigma}_c \in \mathbb{R}^{d\times d}$ of these features

$$\boldsymbol{\mu}_c = \frac{1}{\left|\mathcal{D}_t^c\right|} \sum_{\boldsymbol{x}_i \in \mathcal{D}_t^c} \boldsymbol{h}_i^c \tag{12}$$

$$\boldsymbol{\Sigma}_c = \frac{1}{\left|\mathcal{D}_t^c\right|} \sum_{\boldsymbol{x}_i \in \mathcal{D}_t^c} \left(\boldsymbol{h}_i^c - \boldsymbol{\mu}_c\right)\left(\boldsymbol{h}_i^c - \boldsymbol{\mu}_c\right)^\top \tag{13}$$

where $\boldsymbol{h}_i^c = f(\boldsymbol{x}_i)$.

During subsequent incremental sessions, SDR reuses stored means as prototypes to anchor historical class centers (ensuring stability of prior features), while generating pseudo-intermediate features via sampling from the multivariate Gaussian distribution, thereby simulating diversity

$$\tilde{\boldsymbol{h}}_c \sim \mathcal{N}\left(\boldsymbol{\mu}_c, \boldsymbol{\Sigma}_c\right) \quad \forall c \in \bigcup_{k=0}^{t-1} \mathcal{C}_k \tag{14}$$

where $\tilde{\boldsymbol{h}}_c \in \mathbb{R}^d$ denotes the pseudo-intermediate features for the class c. The frozen backbone f ensures that replayed statistical $(\boldsymbol{\mu}_c, \boldsymbol{\Sigma}_c)$ remain unbiased estimates of the historical feature distribution, enabling accurate feature reconstruction without catastrophic forgetting. By jointly replaying deterministic prototypes (representing stability) and diverse pseudo-intermediate features (representing diversity), SDR forces the projection layer to preserve mappings between intermediate features and their corresponding classifier weights for old classes, when learning such mappings for new classes.

The mechanism's key innovation lies not only in replacing raw data storage with lightweight statistical replay, but also in simultaneously mitigating catastrophic forgetting and preventing feature collisions in incremental learning through SDR.

E. Aligned-Distilled Feature Consistency

To enforce geometric alignment between the discriminative output features from the projection layer and the fixed ETF classifier weights according to NC theory, we design the ADFC module, including feature-classifier geometric alignment and cross-session distribution distillation.

1) Feature-Classifier Geometric Alignment: The angular deviations between all output features and their corresponding class weight vectors of the ETF classifier are minimized via a cosine similarity loss

$$\mathcal{L}_{\text{align}} = \frac{1}{|\mathcal{D}_t|} \sum_{(\boldsymbol{x}_i, y_i)\in\mathcal{D}_t} \left(1 - \frac{\boldsymbol{z}_i^\top \boldsymbol{w}_{y_i}}{\|\boldsymbol{z}_i\|_2 \cdot \|\boldsymbol{w}_{y_i}\|_2}\right) + \frac{1}{|\mathcal{M}_t|} \sum_{(\boldsymbol{h}_j, y_j)\in\mathcal{M}_t} \left(1 - \frac{\boldsymbol{z}_j^\top \boldsymbol{w}_{y_j}}{\|\boldsymbol{z}_j\|_2 \cdot \|\boldsymbol{w}_{y_j}\|_2}\right) \tag{15}$$

where $\boldsymbol{x}_i$ is the real sample in the current session, $\boldsymbol{z}_i = g(f(\boldsymbol{x}_i))$, $\boldsymbol{h}_j$ represents either a prototype or pseudo-intermediate feature, $\boldsymbol{z}_j = g(\boldsymbol{h}_j)$, $\mathcal{D}_t$ denotes the training sample set of session t, $\mathcal{M}_t$ represents the set of all prototypes and their corresponding pseudo-intermediate features for previous classes, generated from stored statistical parameters, $\|\cdot\|_2$ denotes the ℓ_2-norm, $\boldsymbol{w}_{y_i} \in \mathbb{R}^d$ and $\boldsymbol{w}_{y_j} \in \mathbb{R}^d$ represent the weight vectors of classes y_i and y_j in the ETF classifier $\mathbf{W}_{\text{ETF}}$. Notably, $\mathcal{M}_t$ will be discarded after session t.

This objective enforces equiangular separation of class centroids (interclass separability) and intraclass variance collapse (intraclass optimality), as prescribed by NC theory. Note that in the base session (session 0), where no prior classes exist and $\mathcal{M}_t = \emptyset$, $\mathcal{L}_{\text{align}}$ reduces to only the first term.

2) Cross-Session Feature Distillation: The distillation loss is applied to constrain the distribution of output features generated by the current projection layer to match the distribution of previous projection layers. This cross-session distribution consistency prevents catastrophic forgetting, thereby preserving the projection layer's ability to map intermediate features to outputs for previously learned classes in incremental sessions. It can be formulated as follows:

$$\mathcal{L}_{\text{KD}}^i\left(z_i^{t-1}, z_i^t\right) = -\frac{1}{d}\sum_{k=1}^{d} \text{softmax}\left(z_i^{t-1}/\tau\right)_k \cdot \log\left(\text{softmax}\left(z_i^t/\tau\right)_k\right) \tag{16}$$

$$\mathcal{L}_{\text{KD}} = \frac{1}{|\mathcal{D}_t|}\sum_{\boldsymbol{x}_i\in\mathcal{D}_t} \mathcal{L}_{\text{KD}}^i\left(z_i^{t-1}, z_i^t\right) + \frac{1}{|\mathcal{M}_t|}\sum_{\boldsymbol{h}_j\in\mathcal{M}_t} \mathcal{L}_{\text{KD}}^j\left(z_j^{t-1}, z_j^t\right) \tag{17}$$

where $\mathcal{D}_t$ denotes the training sample set of session t, $\mathcal{M}_t$ represents the set of all prototypes and their corresponding pseudo-intermediate features for previous classes, generated from stored statistical parameters, τ is the temperature scaling hyperparameter, d denotes the dimensionality of features, g_{t-1} and g_t represent the previous projection layer and the current projection layer, and z_i^{t-1} and z_i^t denote the output features from g_{t-1} and g_t, respectively. Notably, $\mathcal{M}_t$ is discarded after session t.

The total loss function for the ADFC module is

$$\mathcal{L}_{\text{ADFC}} = \mathcal{L}_{\text{align}} + \mathcal{L}_{\text{KD}}. \tag{18}$$

Notably, $\mathcal{L}_{\text{KD}}$ is disabled in the base session due to no historical projection layer.

IV. Experiments

A. Hyperspectral Datasets

In this study, the Indian Pines, Houston 2013, and Pavia University datasets are used for the experiments.

1) Indian Pines: The Indian Pines dataset, acquired in 1992 using the Airborne Visible/Infrared Imaging Spectrometer (AVIRIS), covers a pine forest in Indiana, USA. It consists of 145×145 pixels and 220 spectral bands, of which 20 bands are affected by water absorption, resulting in 200 usable bands for analysis. We select 620 labeled pixels for training and 9671 for testing across 16 land-cover classes.

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TABLE I
NUMBER OF TRAINING AND TESTING SAMPLES PER CLASS FOR THE INDIAN PINES, HOUSTON 2013, AND PAVIA UNIVERSITY DATASETS

Indian Pines			Houston 2013			Pavia University		
Class Name	Train	Test	Class Name	Train	Test	Class Name	Train	Test
Corn-notill	50	1384	Healthy Grass	198	1053	Asphalt	548	6304
Corn-mintill	50	784	Stressed Grass	190	1064	Meadows	540	18146
Corn	50	184	Synthetic Grass	192	505	Gravel	392	1815
Grass-pasture	50	447	Tree	188	1056	Trees	524	2912
Grass-trees	50	697	Soil	186	1056	Metal Sheets	265	1113
Hay-windrowed	50	439	Water	182	143	Bare Soil	5	4572
Soybean-notill	50	918	Residential	196	1072	Bitumen	5	981
Soybean-mintill	50	2418	Commercial	191	1053	Bricks	5	3364
Soybean-clean	50	564	Road	193	1059	Shadows	5	795
Wheat	50	162	Highway	191	1036	—	—	—
Woods	50	1244	Railway	181	1054	—	—	—
Buildings-Grass-Trees-Drives	50	330	Parking Lot 1	5	1041	—	—	—
Stone-Steel-Towers	5	45	Parking Lot 2	5	285	—	—	—
Alfalfa	5	39	Tennis Court	5	247	—	—	—
Grass-pasture-mowed	5	11	Running Track	5	473	—	—	—
Oats	5	5	—	—	—	—	—	—
Total	620	9671	Total	2108	12197	Total	2289	40002

2) Houston 2013: The Houston 2013 dataset, captured by the National Center for Airborne Laser Mapping, covers the University of Houston campus and adjacent urban areas. It contains 349 × 1905 pixels with 144 spectral bands. We select 2108 training samples and 12 197 test samples across 15 distinct land-cover classes.

3) Pavia University: Acquired in 2003 by the Reflective Optics System Imaging Spectrometer (ROSIS-3) over Pavia, Italy, this dataset has 610 × 340 pixels with 115 spectral bands, of which 12 noisy bands were removed, retaining 103 bands. We allocate 2289 training samples and 40 002 test samples for nine distinct land-cover classes.

The class distributions and corresponding training/testing sample numbers for the three datasets are detailed in Table I.

B. Experiment Setting

1) Environment Setting: The computational infrastructure comprises an Intel Xeon Gold 5318Y CPU with 14-GB RAM and NVIDIA A16 GPU (15-GB VRAM), running Ubuntu 20.04 LTS with a Python 3.10 development environment where PyTorch 2.1.1 is implemented using the CUDA 11.8 parallel computing platform accelerated by cuDNN v8 libraries.

2) Parameter Setting: The spectral dimensionality is reduced to 36 through principal component analysis (PCA), with spatial sampling performed using 9 × 9 pixel patches to construct 9 × 9 × 36 spatial–spectral cubes as network inputs. The LSGA [6] architecture is employed as the backbone network. The hyperparameters in (10) require dataset-specific configurations: $\lambda_1 = 0$ for Indian Pines, 0.5 for Houston, and 0.4 for Pavia University. For (11), the configurations are $\lambda_2 = 1$ for Indian Pines, 0.036 for Houston, and 0.02 for Pavia University. The training protocol is implemented with the SGD optimizer, a learning rate of 0.05, a mini-batch size of 128, and 100 training epochs.

3) Task Setting: Under the few-shot lifelong learning framework, all three datasets are uniformly partitioned into one base session and four incremental sessions according to the class sequence specified in Table I. Specifically, the base session for Indian Pines contains the first 12 classes; the base session for Houston 2013 comprises the first 11 classes; and the base session for Pavia University includes the first five classes. The base session employs complete training and test samples for each respective class. Each incremental session sequentially introduces one new class with only five training samples, while evaluation is performed on the full test sets of all encountered classes (both new and previously learned).

4) Evaluation Metrics: A comprehensive assessment strategy is implemented.

1) Classification accuracy is adopted as the metric for individual class performance.
2) For each session, three metrics are employed: overall accuracy (OA), average accuracy (AA), and Kappa coefficient.
3) The knowledge forgetting rate (KFR) at session t quantifies the relative performance decline for classes learned in sessions 0 to $t-1$ upon new-class introduction

$$\mathrm{KFR}_t = \frac{\mathrm{OA}_{t-1}^{\mathcal{H}(t)} - \mathrm{OA}_{t}^{\mathcal{H}(t)}}{\mathrm{OA}_{t-1}^{\mathcal{H}(t)}} \tag{19}$$

where $\mathcal{H}(t)$ contains all classes from sessions 0 to $t-1$, $\mathrm{OA}_{t-1}^{\mathcal{H}(t)}$ and $\mathrm{OA}_{t}^{\mathcal{H}(t)}$ denote the accuracies of $\mathcal{H}(t)$ after session $t-1$ and t, respectively. A higher KFR indicates more severe catastrophic forgetting. The average KFR is

$$\text{Average KFR} = \frac{1}{T-1}\sum_{t=1}^{T-1}\mathrm{KFR}_t \tag{20}$$

where T is the total number of all sessions.
4) We record OA values after each session and compute the standard deviation across sessions to evaluate temporal dynamics

$$\sigma_{\mathrm{OA}} = \sqrt{\frac{1}{T}\sum\nolimits_{t=0}^{T-1}\left(\mathrm{OA}_t - \overline{\mathrm{OA}}\right)^2} \tag{21}$$

where T is the number of all sessions.

The assessment for lifelong learning utilizes two key indicators: the average OA across all sessions and the final session OA (session 4). The average OA quantifies model stability

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TABLE II
EXPERIMENTAL RESULTS (%) UNDER DIFFERENT INITIAL CLASS QUANTITIES IN BASE SESSION

Dataset	Session	2	5	8	11	14
Indian Pines	Final Session OA	22.23	60.89	88.89	93.15	94.96
	Average OA	59.23	79.73	92.06	94.16	95.16
Houston 2013	Final Session OA	17.80	75.57	78.29	86.17	90.38
	Average OA	54.05	85.57	85.69	88.69	90.10
Pavia University	Final Session OA	63.24	79.87	82.90	—	—
	Average OA	81.62	88.87	83.00	—	—

TABLE III
EXPERIMENTAL RESULTS (%) UNDER DIFFERENT CLASS QUANTITIES FOR INCREMENTAL SESSION

Dataset	Session	1	2	4
Indian Pines	Final Session OA	94.82	94.62	94.61
	Average OA	94.82	94.83	94.89
Houston 2013	Final Session OA	86.98	87.15	86.17
	Average OA	88.19	88.23	88.69
Pavia University	Final Session OA	87.34	85.00	79.87
	Average OA	89.41	89.45	88.87

throughout the lifelong learning process, whereas the final session OA reflects the ultimate classification performance on all learned classes upon completion of lifelong learning.

C. Parameter Analysis

1) Spectral Dimension: To investigate the impact of different spectral dimensions on lifelong learning, we conduct comparative experiments by reducing spectral dimensions via PCA to 16, 28, 36, 48, and 56. As shown in Fig. 3.

1) On Indian Pines, both the average OA and the final session OA decrease monotonically with increasing spectral dimensions. When the number of spectral dimensions reaches 36, both metrics stabilize. This stems from information redundancy in this relatively simple classification context, since higher spectral dimensions introduce additional redundant information.
2) On Houston 2013 and Pavia University, both the average OA and the final session OA initially rise and then decline with increasing spectral dimensions, peaking at 36-D. This trend is particularly pronounced in Pavia University. The initial improvement stems from enriched spectral feature information, while the subsequent decline results from excessive noise beyond the optimal spectral dimensionality.

Although using different spectral dimensions per dataset improves performance, a fixed 36-D setting ensures uniformity and enhances the method's practicality.

2) Base Class Quantities: We investigate the impact of initial class quantities in the base session. For Indian Pines and Houston 2013, initial class quantities are set to 2, 5, 8, 11, and 14 classes, while Pavia University (constrained by limited class inventory) employs 2, 5, and 8 classes. All novel classes are introduced simultaneously in one incremental session. As shown in Table II, increasing base classes enhances both the final session OA and average OA in Indian Pines and Houston 2013, where larger base sets yield more transferable representations while reducing learning complexity in the incremental session. For Pavia University, although the final session OA increases with additional base classes, the average OA declines slightly at eight classes compared with five classes due to reduced base session accuracy caused by interclass confusion.

3) Incremental Class Quantities: As demonstrated in Table III, our method robustly handles these scenarios: 1) on Indian Pines and Houston 2013, the overall performance remains stable when introducing 1, 2, or 4 new classes per incremental session, demonstrating the effectiveness of our proposed method in handling multiple new classes introduced in each session and 2) for Pavia University, performance remains stable with 1–2 new classes per session. However, introducing four new classes causes a significant OA decline in the final session. We attribute this primarily to insufficient training samples for multiclass simultaneous learning and feature confusion due to overlapping spectral characteristics.

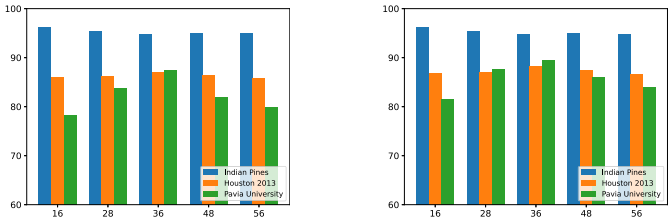


Fig. 3. OA under the different spectral dimensions. (a) Final session OA (%) and (b) average OA (%).

4) Hyperparameter Sensitivity in DBCO: We conduct a comprehensive analysis of the hyperparameter λ_1 in (10), where λ_1 is a nonnegative hyperparameter controlling the trade-off between self-supervised and SupCon learning branches. The mathematically feasible range $\lambda_1 \in [0, 1]$ is systematically evaluated with $\lambda_1 \in \{0, 0.2, 0.4, 0.5, 0.7, 0.9, 1.0\}$. As shown in Fig. 4, the final session OA and average OA exhibit dataset-dependent patterns.

1) On Indian Pines, performance decreases monotonically with increasing λ_1, confirming supervised learning dominance. When $\lambda_1 \geq 0.7$, the model fails to converge.
2) On Houston 2013, performance shows a unimodal distribution where both OA metrics peak at $\lambda_1 = 0.5$, revealing a complementary equilibrium between the two branches.
3) At Pavia University, as λ_1 increases from 0 to 1, both OA metrics initially rise to a peak at $\lambda_1 = 0.4$. Notably, further increasing λ_1 toward 1 yields an unexpected OA rebound.

We experimentally determine the optimal λ_2 in (11) across three hyperspectral datasets. As shown in Fig. 5, only data points with OA > 0.60 are displayed in the bar chart. The experimental results identify optimal λ_2 values as 0.036 for Houston 2013, and 0.02 for Pavia University. The performance remains relatively stable in Indian Pines. We set λ_2 to 1.0.

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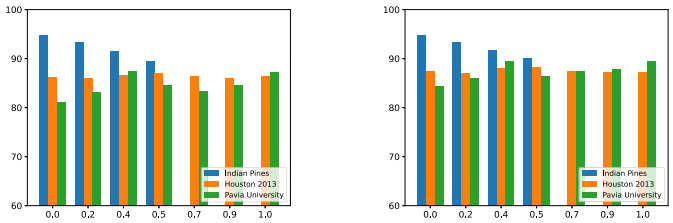


Fig. 4. OA under different hyperparameters λ_1 in (10). (a) Final session OA (%) and (b) average OA (%).

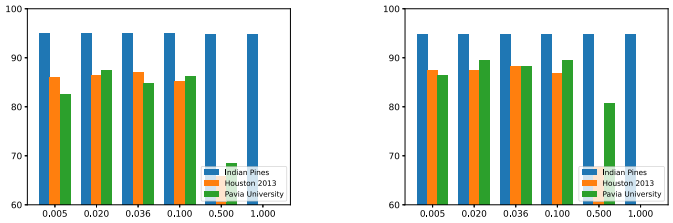


Fig. 5. OA under different hyperparameters λ_2 in (11). (a) Final session OA (%) and (b) average OA (%).

D. Ablation Study

To verify the effectiveness of each module in our proposed method, ablation studies are conducted on Indian Pines, Houston 2013, and Pavia University. As shown in Table IV, experimental results demonstrate that individual module extensions consistently improve classification performance.

1) After integrating the DBCO module into the baseline, the average OA increases by 0.98, 0.70, and 4.85 percentage points to 92.73%, 82.95%, and 73.93% for Indian Pines, Houston 2013, and Pavia University, respectively. The final session OA shows a consistent upward trend, validating the module's effectiveness in enhancing feature representation of the backbone network.
2) Incorporating the SDR mechanism into the baseline results in average OA increases of 1.31, 4.03, and 5.76 percentage points, reaching 93.06%, 86.28%, and 74.84% on the three datasets, respectively. The results demonstrate its effectiveness in lifelong learning, especially in Houston 2013 and Pavia University.
3) Adding the ADFC module to the baseline increases the average OA by 1.36, 2.08, and 7.07 percentage points to 93.11%, 84.33%, and 76.15%, respectively. The final session OA also shows synchronous improvement, confirming its effectiveness in enhancing lifelong learning.
4) When all modules are combined, they collaboratively achieve optimal performance. Compared with the baseline method, the full method increases average OA by 3.07, 5.94, and 20.33 percentage points to 94.82%, 88.19%, and 89.41% across the three datasets, respectively. The final session OA improves by 4.79, 8.01, and 17.40 percentage points to 94.82%, 86.98%, and 87.34%.

TABLE IV
ABLATION STUDY RESULTS (%)

Dataset	Setting	Final Session OA	Average OA
Indian Pines	Baseline	90.03±1.15	91.75±0.73
	Baseline+DBCO	91.05±0.63	92.73±0.73
	Baseline+SDR	92.62±0.68	93.06±0.73
	Baseline+ADFC	92.14±0.69	93.11±0.73
	Baseline+DBCO+SDR	93.62±0.64	93.89±0.66
	Baseline+DBCO+ADFC	93.24±0.42	94.18±0.49
	Baseline+SDR+ADFC	93.72±0.63	93.79±0.61
	Full Method	**94.82±0.49**	**94.82±0.50**
Houston 2013	Baseline	78.97±1.86	82.25±1.64
	Baseline+DBCO	80.00±0.84	82.95±0.98
	Baseline+SDR	84.83±1.54	86.28±1.35
	Baseline+ADFC	81.35±0.93	84.33±1.02
	Baseline+DBCO+SDR	85.35±0.63	86.80±0.72
	Baseline+DBCO+ADFC	81.29±1.18	84.29±0.92
	Baseline+SDR+ADFC	86.26±1.43	87.81±1.01
	Full Method	**86.98±1.95**	**88.19±1.21**
Pavia University	Baseline	69.94±3.08	69.08±1.62
	Baseline+DBCO	71.02±8.23	73.93±7.19
	Baseline+SDR	75.46±2.45	74.84±4.16
	Baseline+ADFC	72.43±5.01	76.15±5.03
	Baseline+DBCO+SDR	78.60±5.98	81.62±6.37
	Baseline+DBCO+ADFC	76.32±7.71	82.80±5.84
	Baseline+SDR+ADFC	78.99±0.93	81.74±2.39
	Full Method	**87.34±2.90**	**89.41±2.59**

TABLE V
CLASSIFICATION RESULTS (%) OF OUR METHOD FOR EACH SESSION ON THE INDIAN PINES DATASET

No.	Class Name	Session 0	Session 1	Session 2	Session 3	Session 4
0	Corn-notill	94.31±0.58	93.76±0.64	93.31±0.78	93.60±1.20	94.21±0.43
1	Corn-mintill	98.90±0.57	99.29±0.37	99.26±0.35	99.34±0.33	97.65±0.87
2	Corn	100.00±0.00	100.00±0.00	100.00±0.00	100.00±0.00	100.00±0.00
3	Grass-pasture	97.09±0.40	97.09±0.28	97.14±0.30	90.65±4.07	94.18±1.21
4	Grass-trees	99.57±0.48	99.57±0.54	99.54±0.85	99.80±0.28	99.86±0.13
5	Hay-windrowed	100.00±0.00	100.00±0.00	99.82±0.09	99.86±0.11	99.82±0.09
6	Soybean-notill	93.68±1.83	92.40±1.26	93.16±1.33	93.51±0.60	94.12±1.67
7	Soybean-mintill	89.21±1.96	89.85±2.20	89.36±1.41	88.68±1.46	89.32±1.92
8	Soybean-clean	95.07±1.00	90.39±2.15	90.50±1.26	90.57±1.43	91.52±1.92
9	Wheat	100.00±0.00	100.00±0.00	100.00±0.00	100.00±0.00	100.00±0.00
10	Woods	99.12±0.10	99.05±0.19	99.12±0.15	99.02±0.06	99.04±0.05
11	Buildings-Grass-Trees-Drives	99.76±0.35	99.94±0.12	99.94±0.12	99.94±0.12	99.88±0.15
12	Stone-Steel-Towers	—	93.33±6.29	97.78±2.81	96.44±3.61	96.89±3.87
13	Alfalfa	—	—	96.41±2.05	98.97±1.26	98.97±1.26
14	Grass-pasture-mowed	—	—	—	100.00±0.00	100.00±0.00
15	Oats	—	—	—	—	100.00±0.00
	OA	95.17±0.62	94.88±0.63	94.80±0.51	94.43±0.33	94.82±0.49
	AA	97.23±0.30	96.51±0.43	96.81±0.33	96.69±0.32	97.22±0.15
	Kappa	94.53±0.70	94.13±0.72	94.05±0.59	93.63±0.37	94.08±0.56

TABLE VI
CLASSIFICATION RESULTS (%) OF OUR METHOD FOR EACH SESSION ON THE HOUSTON 2013 DATASET

No.	Class Name	Session 0	Session 1	Session 2	Session 3	Session 4
0	Healthy Grass	81.18±0.07	81.60±0.18	81.63±0.18	81.44±0.32	81.44±0.20
1	Stressed Grass	99.47±0.62	99.06±1.04	99.10±0.99	98.85±1.27	99.00±1.00
2	Synthetic Grass	99.52±0.20	99.09±0.55	98.93±0.61	99.37±0.15	99.56±0.08
3	Tree	99.51±0.31	99.11±0.28	98.83±0.49	99.09±0.50	99.07±0.58
4	Soil	100.00±0.00	100.00±0.00	99.81±0.38	100.00±0.00	100.00±0.00
5	Water	95.94±0.28	96.64±1.68	96.50±1.40	97.20±1.77	96.50±1.40
6	Residential	83.56±3.63	86.29±3.69	86.29±3.17	82.87±2.08	85.06±1.84
7	Commercial	85.30±9.28	88.91±5.12	87.14±2.08	81.03±6.50	74.76±12.98
8	Road	86.19±0.81	81.32±4.00	79.23±2.35	78.51±2.46	80.02±2.64
9	Highway	85.50±3.54	97.39±0.84	94.44±2.41	93.71±4.64	93.90±5.15
10	Railway	95.56±1.39	90.80±4.12	91.35±3.67	92.79±3.97	92.79±2.46
11	Parking Lot 1	—	54.01±12.09	59.52±10.28	56.54±9.24	54.70±13.48
12	Parking Lot 2	—	—	53.33±12.21	57.54±11.34	60.49±10.94
13	Tennis Court	—	—	—	97.89±1.61	98.87±1.00
14	Running Track	—	—	—	—	99.62±0.76
	OA	91.22±1.44	88.50±1.28	87.51±0.89	86.74±1.47	86.98±1.95
	AA	91.98±1.26	89.52±1.24	86.62±1.22	86.92±1.78	87.72±1.84
	Kappa	90.25±1.60	87.35±1.41	86.31±0.97	85.52±1.61	85.86±2.11

These results demonstrate that the synergistic effects of DBCO, SDR, and ADFC modules effectively address catastrophic forgetting and few-shot overfitting challenges.

E. Experimental Results

1) Classification Results: In few-shot lifelong learning for HSI classification, our proposed method demonstrates exceptional lifelong classification performance, as shown in

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TABLE VII

CLASSIFICATION RESULTS (%) OF OUR METHOD FOR EACH SESSION ON THE PAVIA UNIVERSITY DATASET

No.	Class Name	Session 0	Session 1	Session 2	Session 3	Session 4
0	Asphalt	98.30±0.70	97.81±0.98	88.96±2.35	87.74±5.20	90.14±5.31
1	Meadows	98.22±0.27	90.41±11.06	87.04±7.05	85.95±9.49	88.77±7.66
2	Gravel	95.63±1.63	94.81±2.32	95.75±2.54	83.11±4.32	81.05±4.09
3	Trees	95.24±1.57	96.92±1.49	96.45±1.85	96.23±2.27	96.75±2.10
4	Metal sheets	100.00±0.00	100.00±0.00	99.98±0.04	99.98±0.04	99.98±0.04
5	Bare Soil	—	50.70±10.55	78.60±20.35	85.74±12.26	77.72±13.98
6	Bitumen	—	—	87.62±5.02	88.73±3.44	89.91±3.44
7	Bricks	—	—	—	74.47±4.80	80.05±5.43
8	Shadows	—	—	—	—	77.51±9.34
OA		97.86±0.17	87.62±5.38	87.93±2.51	86.33±3.69	87.34±2.90
AA		97.48±0.26	88.44±1.93	90.63±2.47	87.74±1.09	86.87±1.73
Kappa		96.36±0.28	81.66±7.11	82.82±3.37	81.82±4.45	83.15±3.57

Tables V–VII. For Indian Pines, despite sequentially introducing four incremental sessions from sessions 0 to 4, the OA remains stable between 94.43% and 95.17%, while the AA and Kappa coefficient fluctuate slightly within 96.51%–97.23% and 93.63%–94.53%, respectively, validating the strong robustness of our method in few-shot lifelong learning. Base classes such as corn, grass-trees, hay-windrowed, wheat, woods, and buildings-grass-trees-drives consistently maintain over 99% accuracy, clearly demonstrating the effectiveness of the anti-forgetting mechanism. In incremental sessions, newly added classes achieve remarkable performance even with limited training data: grass-pasture-mowed and oats both attain 100% classification accuracy, while stone-steel-towers and Alfalfa achieve accuracies of 96.89% and 98.97% in session 4, indicating the method's capability to mitigate overfitting under few-shot training conditions.

In Houston 2013, the OA declines from 91.22% in session 0 to 86.98% in session 4, while the AA and Kappa coefficient also decrease by 4.26 and 4.39 percentage points, respectively. Some base classes, such as stressed grass, synthetic grass, tree, and soil, maintain high accuracy above 98%, whereas commercial and road demonstrate catastrophic forgetting, with their accuracy decreasing by 10.54 percentage points (from 85.30% in session 0 to 74.76% in session 4) and 6.17 percentage points (from 86.19% to 80.02%), respectively. Among newly added classes in incremental sessions, tennis court and running track achieve exceptional performance with accuracies of 98.87% and 99.62% in session 4, respectively. However, parking lots 1 and 2 consistently underperform across all sessions.

At Pavia University, the OA decreases by 10.52 percentage points from 97.86% in session 0 to 87.34% in session 4, with the AA and Kappa coefficient decreasing by 10.61 and 13.21 percentage points, respectively. Among base classes, highly distinguishable categories such as metal sheets maintain over 99% accuracy across all sessions, while asphalt, meadows, and gravel experience significant accuracy degradation—dropping by 8.16, 9.45, and 14.58 percentage points, respectively. For new classes in incremental sessions, bitumen demonstrates minor improvement from 87.62% to 89.91%, whereas bare soil exhibits unstable learning trajectories, rising from 50.70% to 77.72% with substantial fluctuations, indicating soil-related feature modeling challenges. Bricks and shadows, added in later sessions, achieve final accuracies of 80.05% and 77.51%.

The results demonstrate that HyperDual maintains the classification capability for old classes while learning new classes.

TABLE VIII

KNOWLEDGE FORGETTING ANALYSIS (%)

Dataset	Session 1	Session 2	Session 3	Session 4	Average
Indian Pines	0.31±0.20	0.15±0.14	0.41±0.25	0.00±0.00	0.22±0.08
Houston 2013	0.12±0.23	0.72±0.74	1.24±0.93	0.55±0.63	0.66±0.35
Pavia University	4.88±6.71	1.86±1.44	1.15±1.44	0.04±0.08	1.98±1.92

TABLE IX

TEMPORAL DYNAMICS ANALYSIS (%)

Dataset	Session 0	Session 1	Session 2	Session 3	Session 4	Average	σ_{OA}
Indian Pines	95.17	94.88	94.80	94.43	94.82	94.82	0.26
Houston 2013	91.22	88.50	87.51	86.74	86.98	88.19	1.75
Pavia University	97.86	87.62	87.93	86.33	87.34	89.41	4.55

2) Knowledge Forgetting and Temporal Dynamics Analysis: As shown in Table VIII, Pavia University exhibits severe knowledge forgetting (4.88%) in session 1, with a standard deviation of 6.71, indicating catastrophic forgetting occurred in certain experimental runs. In contrast, Indian Pines demonstrates exceptional stability, with KFR below 0.41% across all sessions and achieving zero forgetting (0.00%) in session 4. For Houston 2013, a KFR peak of 1.24% occurs in session 3. Overall, our method effectively suppresses catastrophic forgetting while maintaining robust classification capabilities for previously learned categories. As shown in Table IX, Pavia University shows the highest volatility while Indian Pines maintains optimal stability ($\sigma_{\mathrm{OA}} = 0.26$).

3) t-SNE Visualizations: To validate that HyperDual maintains interclass separability and intraclass compactness in the feature space during few-shot lifelong learning, we conduct t-SNE [72] visualizations across all sessions. For each class, 100 samples are randomly selected from the combined training and test sets (utilizing all available samples for classes with fewer than 100 instances). As demonstrated in Figs. 6–8, the base sessions exhibit well-defined cluster separation and intraclass compactness, while subsequent incremental sessions consistently preserve these properties for both new and old classes. This structural stability confirms that HyperDual effectively approximates a simplex ETF throughout the learning process.

F. Comparative Experiments

To comprehensively evaluate the effectiveness of our proposed HyperDual in few-shot lifelong learning for HSI classification, we conduct systematic comparisons with NC-FSCIL [33], FeTrIL [24], FeCAM [54], Aper-Finetune [45], OrCo [34], and HyperKD [17]. A brief introduction to the comparative methods is provided below.

1) *NC-FSCIL:* A feature space-based few-shot lifelong learning method that leverages a preassigned ETF classifier to enforce feature alignment, maintaining theoretically optimal feature-classifier geometry during lifelong learning while preventing catastrophic forgetting.
2) *FeTrIL:* A feature space-based method that integrates a fixed feature extractor and a pseudo-features generator.

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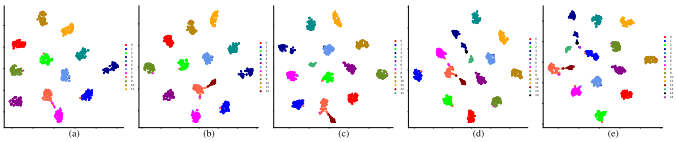


Fig. 6. t-SNE visualization of feature distributions across all sessions in the Indian Pines dataset. (a) Session 0. (b) Session 1. (c) Session 2. (d) Session 3. (e) Session 4.

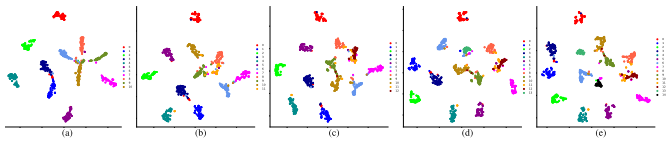


Fig. 7. t-SNE visualization of feature distributions across all sessions in the Houston 2013 dataset. (a) Session 0. (b) Session 1. (c) Session 2. (d) Session 3. (e) Session 4.

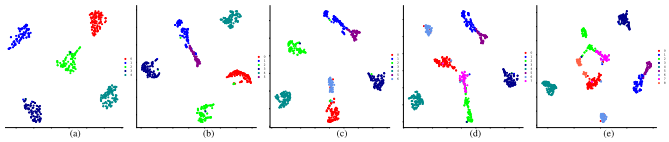


Fig. 8. t-SNE visualization of feature distributions across all sessions in the Pavia University dataset. (a) Session 0. (b) Session 1. (c) Session 2. (d) Session 3. (e) Session 4.

The generator employs geometric translation of new class features to synthesize past-class representations.

3) *FeCAM:* A feature space-based method replacing Euclidean distance with an anisotropic Mahalanobis distance to optimize prototype classifiers. By modeling feature covariance relations, it significantly enhances classification performance under nonstationary data distributions.
4) *Aper-Finetune:* It trains a pretrained model in the base session for strong generalization, while an adaptive model is trained in session 1 to specialize in new-class adaptation. In subsequent sessions, feature outputs from both models are merged to enhance discriminability for novel classes.
5) *OrCo:* A feature space-based method leveraging hybrid contrastive pretraining to enhance generalization in the base session and introducing orthogonal-contrastive (OrCo) loss to overcome data scarcity challenges.
6) *HyperKD:* A distillation-based method that selects and stores the most representative exemplars for subsequent sessions. When learning new classes, it preserves prior knowledge via cross-spectral–spatial distillation.

1) Overall Performance Across Sessions: As shown in Table X for Indian Pines, HyperDual achieves an OA of 95.17% in Session 0, outperforming all baseline methods (the closest competitor: HyperKD at 95.13%). Throughout four incremental sessions, it maintains stable OA values between 94.43% and 94.88%, consistently surpassing other methods. In Session 4, HyperDual attains 94.82% OA, demonstrating a gain of 2.27 percentage points over the second-best method FeCAM (92.55%). The average OA across sessions is 94.82%, exceeding FeCAM's 92.71% by 2.11 percentage points.

In Houston 2013, HyperDual achieves an OA of 91.22% in session 0, significantly outperforming OrCo's 90.10%. Although the OA gradually declines to 86.98% by session 4, its cumulative performance decline (4.24 percentage points) remains substantially smaller than that of all baseline methods (e.g., FeCAM's 8.37 percentage points degradation). In session 4, HyperDual surpasses the second-best method, OrCo by 5.08 percentage points (86.98% versus 81.90%). The average OA across all sessions is 88.19%, outperforming all baseline methods (the closest competitor: FeCAM at 83.67%).

At Pavia University, while HyperDual's initial OA (97.86%) slightly trails OrCo (98.68%) in session 0, it exhibits remarkable incremental adaptability. By session 4, the OA reaches 87.34%, achieving an 11.95 percentage points improvement over FeCAM's 75.39%. The average OA is 89.41%, surpassing FeCAM's 82.63% by 6.78 percentage points.

2) Final Session Classification Performance: The classification results for the final incremental session (session 4) are summarized in Tables XI–XIII.

On Indian Pines, three critical findings emerge. First, HyperDual achieves state-of-the-art performance in 10 out of 16 classes (including ties with other methods), demonstrating exceptional capability in most categories. Second, while HyperDual does not lead in six classes (corn-mintill, hay-windrowed, soybean-clean, woods, buildings-grass-trees-drives, and stone-steel-towers), its performance remains highly competitive. For example, in woods, it reaches 99.04% accuracy, marginally lower than FeCAM's 99.10%. Third, HyperDual achieves 94.82% OA (2.27 percentage points higher than FeCAM's 92.55%), 97.22% AA (0.98 percentage points higher than FeCAM's 96.24%), and 94.08% Kappa (2.57 percentage points higher than FeCAM's 91.51%).

In Houston 2013, HyperDual demonstrates remarkable superiority. First, in overall metrics, HyperDual achieves state-of-the-art performance with 86.98% OA (surpassing the second-best method OrCo by 5.08 percentage points), 87.72% AA (3.59 percentage points higher than OrCo), and 85.86% Kappa (a 5.47 percentage points improvement over OrCo), validating its global optimization capability. Second, in per-class accuracy, HyperDual achieves the highest results in 6 out of 15 classes (including a tie with FeCAM in soil). For instance, it significantly surpasses the second-best method, NC-FSCIL (53.68%), in commercial with 74.76% accuracy and outperforms OrCo (86.80%) by 7.10 percentage points in Highway (93.90%). Notably, even in nonleading categories, HyperDual remains highly competitive. For example, in healthy grass, it attains 81.44% accuracy, closely trailing behind HyperKD's 82.55%.

At Pavia University, first, in overall metrics, our method achieves state-of-the-art results with 87.34% OA (surpassing the second-best FeCAM by 11.95 percentage points), 86.87% AA (7.26 percentage points higher than FeTrIL), and 83.15% Kappa (15.71 percentage points improvement over FeCAM). Second, in per-class accuracy, our method attains the highest accuracy in 2 out of 9 classes. Notably, it significantly outperforms existing methods in challenging categories such

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TABLE X

COMPARATIVE EXPERIMENTAL RESULTS (%) ON THE INDIAN PINES, HOUSTON 2013, AND PAVIA UNIVERSITY DATASETS

Dataset	Session	NC-FSCIL [33]	FeTrIL [24]	FeCAM [54]	Aper-Finetune [45]	OrCo [34]	HyperKD [17]	Ours
Indian Pines	0	93.55±0.81	94.45±0.76	92.97±0.37	92.70±0.38	93.66±0.42	95.13±0.85	**95.17±0.62**
	1	92.91±1.07	89.98±4.53	92.71±0.34	80.65±6.65	92.67±0.52	93.11±1.35	**94.88±0.63**
	2	91.10±1.24	92.73±0.81	92.70±0.34	73.91±11.99	92.22±0.55	91.46±2.09	**94.80±0.51**
	3	91.17±0.73	91.58±0.71	92.63±0.36	64.40±14.77	91.74±0.75	90.14±1.41	**94.43±0.33**
	4	90.03±1.15	90.97±0.71	92.55±0.37	54.07±15.39	90.93±1.08	89.83±1.45	**94.82±0.49**
	Average	91.75±0.73	91.94±1.10	92.71±0.35	73.15±9.50	92.24±0.55	91.93±1.24	**94.82±0.50**
Houston 2013	0	89.47±1.53	87.10±1.46	89.28±0.79	88.72±0.52	90.10±1.32	82.08±4.36	**91.22±1.44**
	1	84.99±1.73	83.10±2.95	84.86±1.28	73.12±4.44	82.04±5.18	79.88±1.72	**88.50±1.28**
	2	79.38±1.81	79.35±3.31	82.17±2.00	70.15±4.07	79.02±5.45	76.60±2.27	**87.51±0.89**
	3	78.46±2.35	78.25±2.81	81.13±2.56	61.73±12.63	81.73±1.49	76.68±4.13	**86.74±1.47**
	4	78.97±1.86	79.46±2.76	80.91±2.20	62.12±12.22	81.90±1.65	76.54±3.27	**86.98±1.95**
	Average	82.25±1.64	81.45±2.40	83.67±1.74	71.17±6.40	82.96±2.38	78.36±1.55	**88.19±1.21**
Pavia University	0	98.48±0.28	98.27±0.28	98.34±0.29	98.24±0.28	**98.68±0.25**	98.11±0.49	97.86±0.17
	1	45.19±0.58	79.90±5.14	83.15±3.52	56.29±8.57	46.27±0.39	79.70±4.61	**87.62±5.38**
	2	64.73±3.83	75.02±5.85	80.56±4.26	40.51±8.06	41.85±9.33	73.89±5.57	**87.93±2.51**
	3	67.08±2.90	73.54±3.69	75.72±3.85	41.45±7.92	47.43±8.94	71.85±5.11	**86.33±3.69**
	4	69.94±3.08	75.01±3.24	75.39±3.76	41.99±7.98	51.20±9.15	67.59±6.64	**87.34±2.90**
	Average	69.08±1.62	80.35±3.21	82.63±3.02	55.70±6.46	57.09±4.73	78.23±4.06	**89.41±2.59**

TABLE XI

COMPARATIVE EXPERIMENTAL RESULTS (%) OF THE FINAL SESSION ON THE INDIAN PINES DATASET

Class No.	Class Name	NC-FSCIL [33]	FeTrIL [24]	FeCAM [54]	Aper-Finetune [45]	OrCo [34]	HyperKD [17]	Ours
0	Corn-notill	91.21±2.34	87.21±1.46	92.08±0.81	40.14±21.81	93.02±1.06	90.16±3.56	**94.21±0.43**
1	Corn-mintill	81.91±16.38	95.03±2.23	**97.68±1.02**	20.61±15.19	85.94±7.31	95.23±3.63	97.65±0.87
2	Corn	**100.00±0.00**	98.48±0.41	**100.00±0.00**	77.72±23.22	99.35±1.05	96.20±2.77	**100.00±0.00**
3	Grass-pasture	69.89±2.23	70.02±2.40	94.09±0.76	66.09±8.98	86.62±8.83	78.43±3.75	**94.18±1.21**
4	Grass-trees	99.05±0.97	97.39±1.35	99.14±0.26	71.74±20.77	99.08±0.78	97.96±1.06	**99.86±0.13**
5	Hay-windrowed	99.95±0.09	**100.00±0.00**	99.73±0.44	83.33±12.17	99.86±0.11	99.82±0.36	99.82±0.09
6	Soybean-notill	88.41±2.09	93.27±3.09	88.74±1.75	57.93±10.18	88.61±2.58	88.82±3.56	**94.12±1.67**
7	Soybean-mintill	84.14±1.45	84.84±2.73	83.61±0.61	30.26±23.33	82.38±3.22	83.03±4.20	**89.32±1.92**
8	Soybean-clean	91.21±1.29	**92.09±0.81**	91.95±0.57	37.94±25.72	90.25±0.71	80.71±7.74	91.52±1.92
9	Wheat	**100.00±0.00**	**100.00±0.00**	**100.00±0.00**	97.16±2.01	**100.00±0.00**	**100.00±0.00**	**100.00±0.00**
10	Woods	98.63±0.30	98.31±0.48	**99.10±0.15**	97.59±1.93	98.89±0.16	95.53±1.97	99.04±0.05
11	Buildings-Grass-Trees-Drives	99.76±0.48	98.48±1.57	**100.00±0.00**	78.97±19.89	99.88±0.24	96.73±2.72	99.88±0.15
12	Stone-Steel-Towers	98.22±2.59	97.33±0.89	97.33±4.31	96.89±4.12	**100.00±0.00**	98.22±2.59	96.89±3.87
13	Alfalfa	95.90±3.08	95.38±1.03	96.41±1.26	97.44±0.00	93.85±4.76	95.90±4.17	**98.97±1.26**
14	Grass-pasture-mowed	**100.00±0.00**	**100.00±0.00**	**100.00±0.00**	**100.00±0.00**	**100.00±0.00**	**100.00±0.00**	**100.00±0.00**
15	Oats	**100.00±0.00**	**100.00±0.00**	**100.00±0.00**	**100.00±0.00**	**100.00±0.00**	80.00±40.00	**100.00±0.00**
	OA	90.03±1.15	90.97±0.71	92.55±0.37	54.07±15.39	90.93±1.08	89.83±1.45	**94.82±0.49**
	AA	93.64±1.23	94.24±0.21	96.24±0.37	72.11±10.46	94.86±0.89	92.30±1.94	**97.22±0.15**
	Kappa	88.65±1.30	89.74±0.80	91.51±0.42	51.46±15.19	89.68±1.22	88.41±1.63	**94.08±0.56**

TABLE XII

COMPARATIVE EXPERIMENTAL RESULTS (%) OF THE FINAL SESSION ON THE HOUSTON 2013 DATASET

Class No.	Class Name	NC-FSCIL [33]	FeTrIL [24]	FeCAM [54]	Aper-Finetune [45]	OrCo [34]	HyperKD [17]	Ours
0	Healthy Grass	81.60±0.26	81.52±0.29	81.50±0.24	71.13±15.87	81.77±0.15	**82.55±1.68**	81.44±0.20
1	Stressed Grass	96.84±4.71	97.12±1.82	**99.25±0.73**	75.92±32.64	98.50±0.78	92.35±4.99	99.00±1.00
2	Synthetic Grass	93.74±2.12	94.61±1.63	95.80±1.09	92.16±2.95	98.10±0.54	97.15±0.91	**99.56±0.08**
3	Tree	**99.36±0.53**	96.69±0.50	98.18±0.88	68.88±31.45	99.15±0.55	92.75±4.28	99.07±0.58
4	Soil	99.75±0.40	99.96±0.08	**100.00±0.00**	88.47±17.00	99.92±0.15	95.95±2.13	**100.00±0.00**
5	Water	95.38±5.69	93.43±1.86	95.24±1.12	94.55±1.56	**98.18±2.01**	93.15±4.71	96.50±1.40
6	Residential	81.25±4.04	72.67±11.70	82.74±4.46	37.54±15.48	76.72±8.94	60.63±18.81	**85.06±1.84**
7	Commercial	53.68±5.09	52.40±4.20	44.35±9.87	38.25±8.01	39.64±15.27	46.04±9.41	**74.76±12.98**
8	Road	70.86±3.23	74.58±2.84	**80.91±4.45**	57.37±11.67	76.19±3.89	77.15±10.67	80.02±2.64
9	Highway	64.09±20.61	69.34±13.16	72.93±8.47	51.37±22.10	86.80±12.90	71.33±14.21	**93.90±5.15**
10	Railway	87.95±5.56	89.79±2.07	89.66±7.69	51.97±14.03	76.09±14.75	56.64±16.34	**92.79±2.46**
11	Parking Lot 1	31.68±10.74	39.79±13.13	40.69±13.45	29.82±8.63	**66.36±14.20**	64.86±17.77	54.70±13.48
12	Parking Lot 2	73.40±17.95	76.49±3.54	65.89±12.41	**82.39±10.09**	66.53±6.30	73.05±9.37	60.49±10.94
13	Tennis Court	97.41±1.46	89.55±13.35	99.68±0.40	**99.76±0.32**	98.46±1.36	97.09±3.15	98.87±1.00
14	Running Track	99.96±0.08	99.92±0.17	**100.00±0.00**	**100.00±0.00**	99.62±0.28	96.58±4.09	99.62±0.76
	OA	78.97±1.86	79.46±2.76	80.91±2.20	62.12±12.22	81.90±1.65	76.54±3.27	**86.98±1.95**
	AA	81.80±2.00	81.86±2.18	83.12±1.66	69.30±9.78	84.13±1.36	79.82±2.65	**87.72±1.84**
	Kappa	77.25±1.99	77.79±2.97	79.36±2.35	59.67±12.51	80.39±1.78	74.63±3.51	**85.86±2.11**

as asphalt (90.14% versus FeCAM's 78.01%) and meadows (88.77% versus FeCAM's 82.32%). Third, while not leading in all classes, our method remains competitive. For example, in metal sheets, it achieves 99.98% accuracy, closely trailing FeCAM and Aper-Finetune (both 100.00%), and in trees, it attains 96.75% accuracy, slightly below OrCo's 97.17%.

The comprehensive analysis demonstrates that HyperDual establishes new state-of-the-art performance in both base and incremental sessions for few-shot lifelong HSI classification. In addition, in the final session, HyperDual achieves optimal classification accuracy for the majority of land-cover categories while maintaining highly competitive performance

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TABLE XIII
COMPARATIVE EXPERIMENTAL RESULTS (%) OF THE FINAL SESSION ON THE PAVIA UNIVERSITY DATASET

Class No.	Class Name	NC-FSCIL [33]	FeTrIL [24]	FeCAM [54]	Aper-Finetune [45]	OrCo [34]	HyperKD [17]	Ours
0	Asphalt	36.82±18.73	70.86±3.65	78.01±8.96	0.96±1.16	8.24±16.45	65.52±8.09	**90.14±5.31**
1	Meadows	82.00±10.17	76.80±9.15	82.32±13.04	20.28±16.59	37.37±20.46	57.64±18.89	**88.77±7.66**
2	Gravel	63.06±14.86	69.33±7.72	**83.28±2.91**	70.03±12.07	44.41±36.29	80.08±6.25	81.05±4.09
3	Trees	88.71±4.65	89.58±2.87	94.79±1.17	96.33±1.58	**97.17±1.25**	85.08±9.34	96.75±2.10
4	Metal Sheets	99.34±0.41	94.59±3.63	**100.00±0.00**	**100.00±0.00**	99.95±0.11	99.89±0.14	99.98±0.04
5	Bare Soil	44.66±16.79	53.92±16.90	49.43±25.28	**91.70±3.52**	87.00±14.25	88.44±18.21	77.72±13.98
6	Bitumen	96.33±1.44	85.36±7.20	91.29±1.89	**98.29±0.28**	90.50±8.52	86.91±8.49	89.91±3.44
7	Bricks	66.19±18.53	77.10±7.64	36.78±5.79	62.28±12.89	**84.03±8.07**	58.81±14.43	80.05±5.43
8	Shadows	91.72±7.49	**98.97±0.60**	65.69±13.54	77.64±21.50	93.08±5.24	66.74±18.73	77.51±9.34
	OA	69.94±3.08	75.01±3.24	75.39±3.76	41.99±7.98	51.20±9.15	67.59±6.64	**87.34±2.90**
	AA	74.31±1.89	79.61±0.85	75.73±0.44	68.61±3.94	71.31±5.02	76.57±2.80	**86.87±1.73**
	Kappa	60.57±3.42	67.34±3.71	67.44±3.75	34.71±7.18	42.27±9.47	60.14±6.72	**83.15±3.57**

TABLE XIV
PARAMETERS (K), MMAC, AND INFERENCE LATENCY (MS/SAMPLE) ACROSS METHODS

Computational Metrics	NC-FSCIL [33]	FeTrIL [24]	FeCAM [54]	Aper-Finetune [45]	OrCo [34]	HyperKD [17]	Ours
Parameters (K)	665.40	536.30	536.28	1072.56	594.24	**497.30**	665.40
Computation Cost (MMac)	37.00	**36.87**	**36.87**	73.75	36.93	36.88	37.00
Inference Latency (ms/sample)	2.03	**1.77**	2.16	3.62	2.44	1.87	2.04

in nondominant categories. Collectively, extensive experiments across three datasets validate that HyperDual not only addresses catastrophic forgetting but also resists few-shot overfitting, ultimately attaining optimal classification accuracy.

3) Computational Complexity: To compare computational complexity across methods, we report the number of parameters, computational cost (MMac), and inference latency on the Indian Pines dataset after the final session (representative for all datasets due to identical architectures). As shown in Table XIV, we observe.

1) Compared with other methods, our method has a parameter count of 665.40K, matching that of the NC-FSCIL method but being lower than Aper-Finetune's 1072.56K, while remaining slightly higher than some other approaches.
2) With a computational cost of 37.00 MMac, our method demonstrates comparable efficiency to other approaches. This efficiency makes our method suitable for resource-constrained environments.
3) At 2.04 ms/sample, our method incurs slightly higher latency than some alternatives but remains within the acceptable range, and significantly reduces inference latency compared with Aper-Finetune.

Aper-Finetune's lowest-performing results stem from its requirement to train both a base model during session 0 and an adaptive model in session 1, necessitating computationally intensive execution across all subsequent phases. Our method adds a projection layer, slightly increasing parameters (665.40K) and latency (2.04 ms/sample) compared with lighter-weight alternatives. Despite these minor increases, HyperDual achieves state-of-the-art lifelong learning classification performance, demonstrating robust overall efficacy.

G. Visualize Results

Figs. 9–11 present the classification maps for the final incremental session (session 4).

As shown in Fig. 9 for Indian Pines, all baseline methods exhibit significant confusion between spectrally and spatially similar category pairs such as soybean-notill (brown)/soybean-mintill (dark green) and grass-pasture (yellow)/grass-pasture-mowed (dark gray). In the bottom region, NC-FSCIL and FeTrIL both misclassify grass-pasture (yellow) as grass-pasture-mowed (dark gray), while FeCAM, OrCo, and HyperKD predominantly misclassify soybean-notill (brown) as soybean-mintill (dark green). For the soybean-mintill (dark green) region in the upper-right quadrant, all baselines demonstrate varying degrees of misclassification. Among them, Aper-Finetune exhibits the least accurate classification visualization due to extensive misclassifications. In contrast, despite minor classification errors in the upper-right soybean-mintill (dark green) region, our method achieves the lowest overall noise contamination among all methods, resulting in visualizations closest to the ground truth.

As shown in Fig. 10 for Houston 2013, (top) full-scene ground truth reference, and (bottom) provides a comparison of the red-boxed region. FeTrIL, NC-FSCIL, FeCAM, Aper-Finetune, and OrCo all generate green-toned misclassified pixels along inner playground edges, specifically misclassifying synthetic grass (blue) as other categories. In addition, FeTrIL exhibits additional residential (orange) misclassifications. HyperKD mislabels stressed grass (green) as healthy grass (red) in the bottom-left quadrant, mislabels running track (dark gray) as tennis court (dark green) along the playground periphery, and generates residential (orange) labeling errors along inner boundaries. In contrast, our method demonstrates near-perfect alignment with the ground truth.

As shown in Fig. 11, NC-FSCIL introduces patchy meadows (green) and gravel (blue) noisy regions in central bare soil (cyan), while misclassifying right-side asphalt (red) as bitumen (brown). FeTrIL exhibits widespread meadows (green) noise contamination in central bare soil (cyan), with bare soil (cyan) misclassifications in the bottom meadows (green). FeCAM contains meadows/gravel misclassifications in central bare

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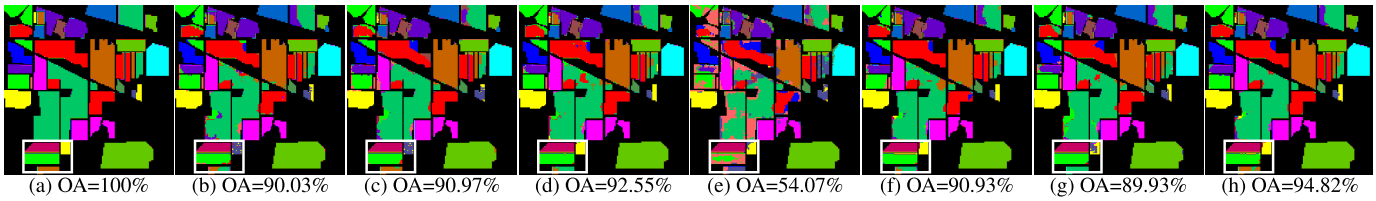


Fig. 9. Classification maps for the final session (session 4) in the Indian Pines dataset. A white bounding box is annotated on each subfigure to highlight a key region for comparison. (a) Ground truth. (b) NC-FSCIL. (c) FeTrIL. (d) FeCAM. (e) Aper-Finetune. (f) OrCo. (g) HyperKD. (h) Ours.

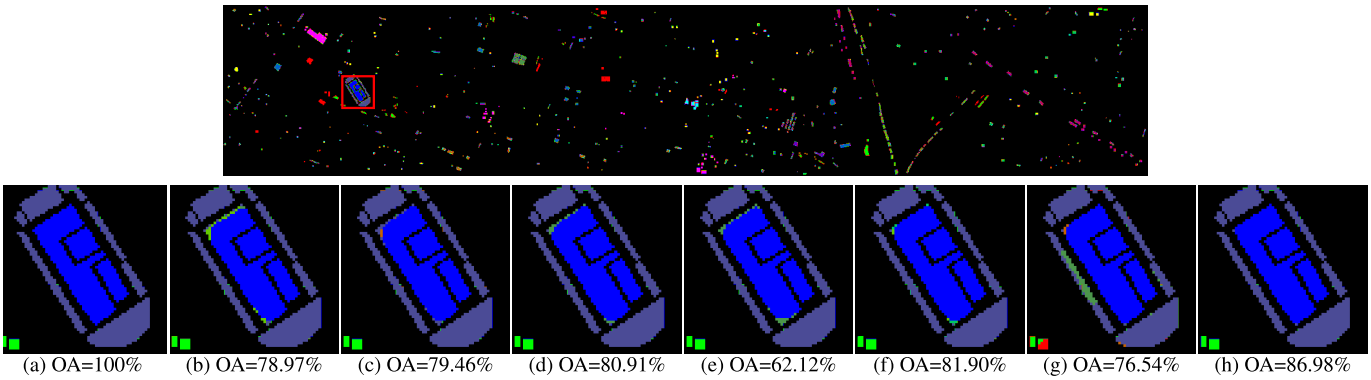


Fig. 10. Classification maps for the final session (session 4) in the Houston 2013 dataset. (Top) Full-scene ground truth reference and (bottom) provide a comparison of the red-boxed region. (a) Ground truth. (b) NC-FSCIL. (c) FeTrIL. (d) FeCAM. (e) Aper-Finetune. (f) OrCo. (g) HyperKD. (h) Ours.

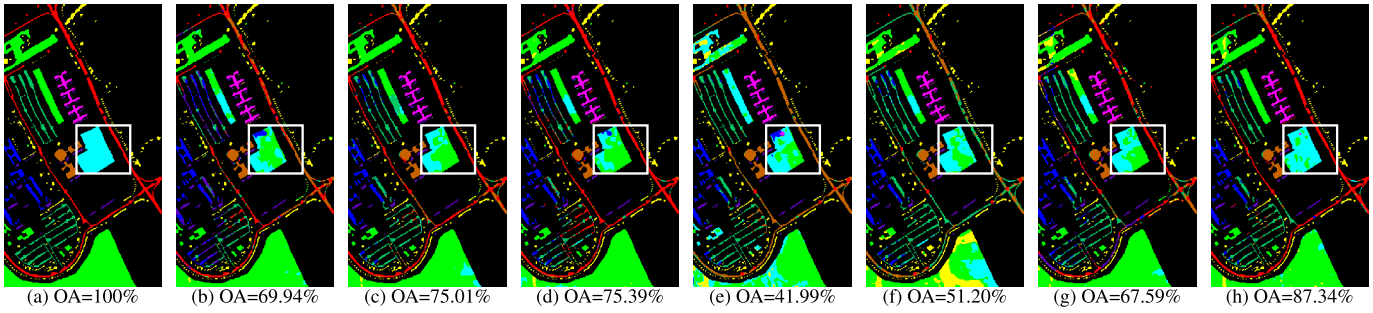


Fig. 11. Classification maps for the final session (session 4) in the Pavia University dataset. A white bounding box is annotated on each subfigure to highlight a key region for comparison. (a) Ground truth. (b) NC-FSCIL. (c) FeTrIL. (d) FeCAM. (e) Aper-Finetune. (f) OrCo. (g) HyperKD. (h) Ours.

soil (cyan) with sporadic trees (yellow) errors in the bottom meadows (green). Aper-Finetune introduces numerous bare soil (cyan) and trees (yellow) noise patches in the bottom meadows (green), while contaminating the central bare soil (cyan) region with meadows (green), gravel (blue), and metal sheets (pink) noise. It also systematically mislabels right-side asphalt (red) as bitumen (brown). OrCo similarly introduces meadows (green) noise in central bare soil (cyan), injects large-scale bare soil (cyan) and trees (yellow) noise in bottom meadows (green), and simultaneously misclassifies right-side asphalt (red) as both bitumen (brown) and Bricks (dark green). HyperKD introduces minor bare soil (cyan) and trees (yellow) noise in the bottom meadows (green), while showing significant meadows (green) contamination in central bare soil (cyan). Our method induces minimal noise in central and bottom areas, while achieving the lowest contamination and fewest errors among all methods.

Visual analysis shows that HyperDual aligns most closely with the ground truth. It significantly reduces confusion among similar land-cover categories while mitigating catastrophic forgetting and overfitting, achieving the highest accuracy.

V. Conclusion

In this study, we develop HyperDual, an integrative framework that systematically coordinates feature optimization, knowledge preservation, and cross-task alignment through three synergistic components. The proposed DBCO module strengthens feature discriminability across both base and novel classes. Through the replay of prototype features and pseudo-intermediate features, the SDR mechanism provides an effective safeguard against catastrophic forgetting with strict privacy compliance without requiring raw data storage. The ADFC module bridges theoretical insights from NC with practical learning stability by enforcing geometric alignment with the ETF classifier while preserving feature distribution continuity.

Experimental validation across three HSI datasets confirms HyperDual's consistent superiority in balancing knowledge retention and novel-class adaptation. The framework pioneers

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theoretically grounded few-shot lifelong learning systems for multimodal remote sensing (e.g., classifying emerging disaster zones under sample scarcity [73]), with direct extensions enabling privacy-preserving continuous diagnosis of rare diseases in medical imaging [74].

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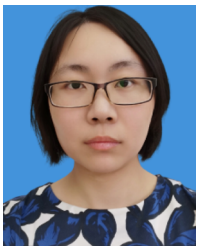
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